

ID6108- Course project

Optimization of Aircraft bearing bracket: Topology and Lattice structure approaches

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Objectives

❑ Stress distribution analysis:

- ✓ By performing static stress simulations on the original, topology-optimized CAD models.

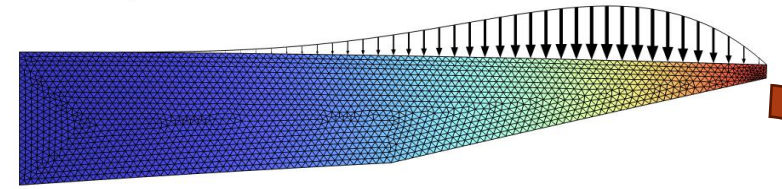
❑ Mass reduction of the component:

- ✓ By doing Shape and Topology optimization.

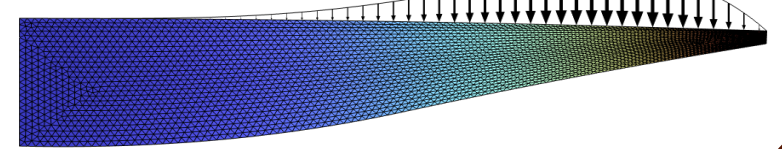
❑ Design optimization:

- ✓ Explore and compare the Shape optimization, Topology optimization, and the integration of lattice structures.

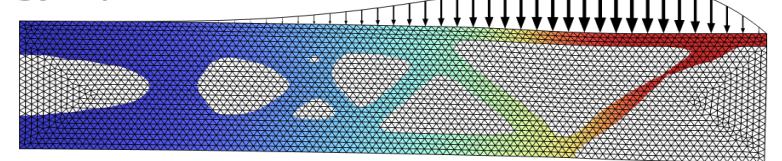
Parameter Optimization



Shape Optimization



Topology Optimization



Introduction

❑ Aircraft bearing Brackets & Their Importance:

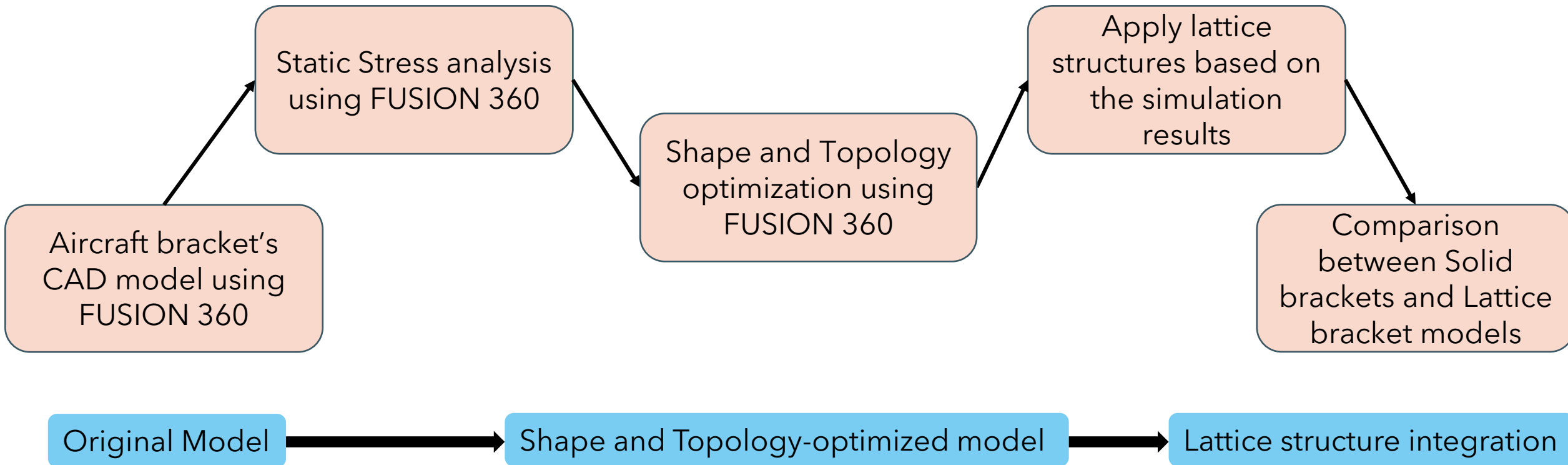
- ✓ Structural components used to mount/support subsystems (e.g., actuators, avionics, landing gear).
- ✓ Must handle loads, vibration, and fatigue.
- ✓ **Weight savings** directly impact **fuel efficiency, payload, and cost.**

❑ Why Additive manufacturing?

- ✓ Goal: Reduce bracket weight without sacrificing structural integrity.
- ✓ Solid designs often contain non-load-bearing material.
- ✓ Use of **shape and topology optimization** helps remove excess mass.
- ✓ Benefits: 40-60% weight reduction, part consolidation, customization.



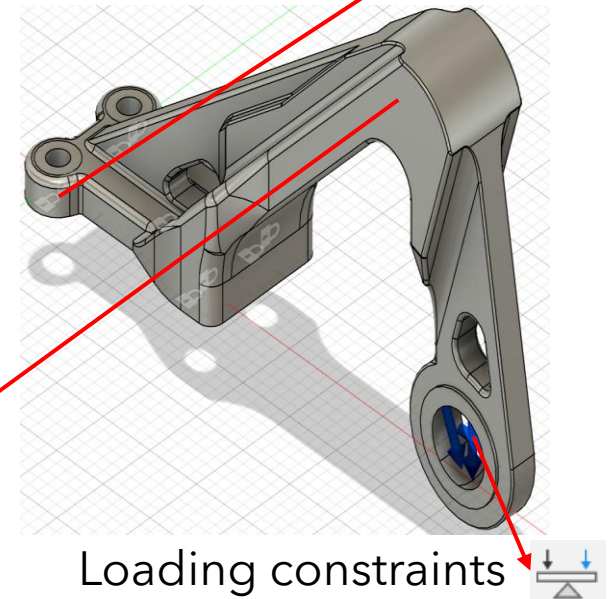
Project Workflow



Simulation setup

Software used	Autodesk Fusion 360
Bracket material	17-4 PH stainless steel powder
Analysis types	Static stress, Shape & Topology optimization
Loading constraints	Load (fx): 250N horizontal Load (fy): 500N vertical Load : 375N at 45° angle
Lattice structures used	Gyroid, X-Cell, Schwarz-P, Schwarz-D
Solidity in lattices	50%
Lattice structure Cell size	2.5 mm

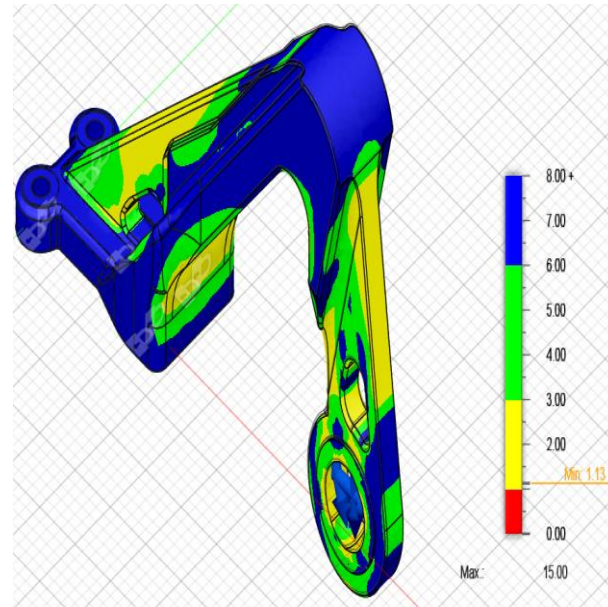
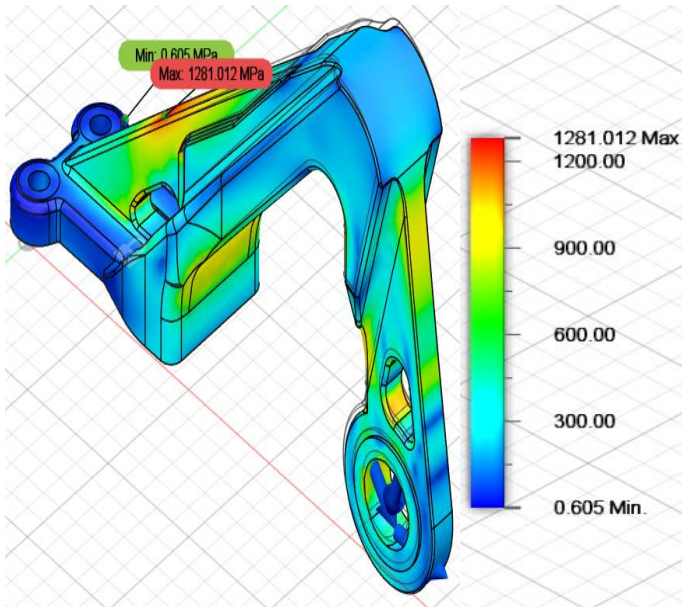
Structural constraints 



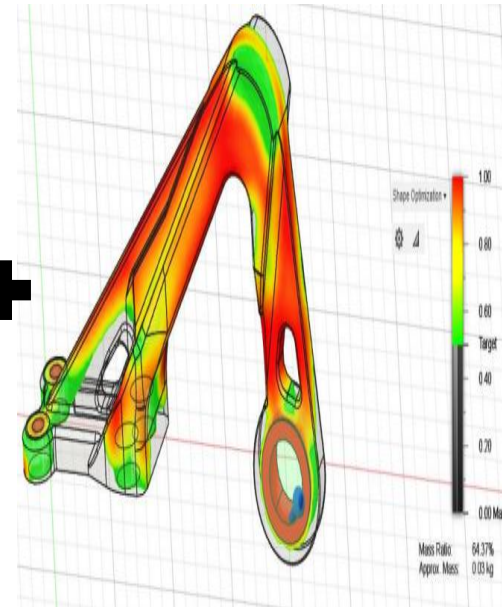
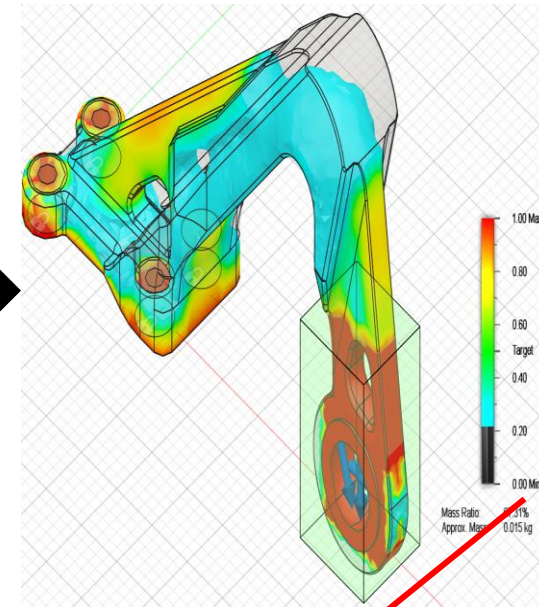
Box Dimensions → Length: 50 mm, Width: 13 mm, Height: 33 mm

Simulation results

Static Stress analysis of the original Aircraft Bracket



Shape Optimization of the bracket



Von-Mises Stress distribution

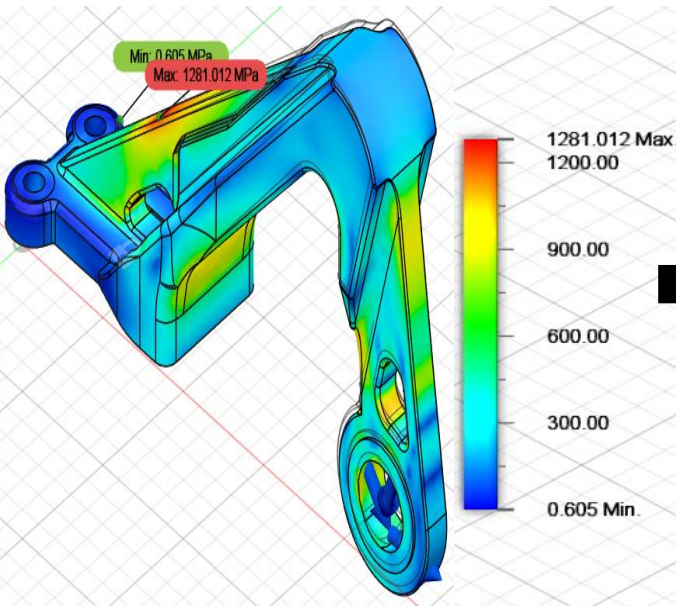
Factor of Safety ~1.13

Mass Reduced to 29% of Original ~15 gms

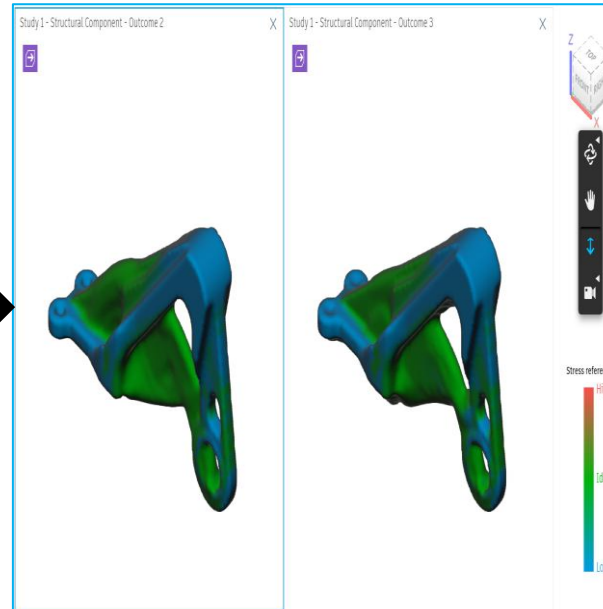
Simulation results

Topology Optimization of the original bracket

Von-Mises Stress distribution for the original model



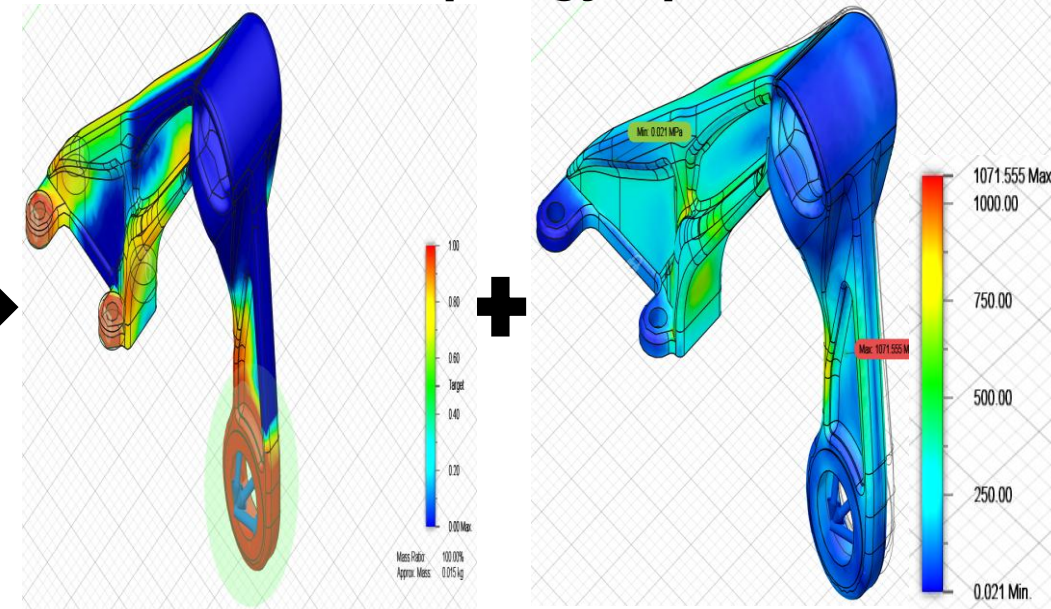
Max. VM Stress~**1281 MPa**



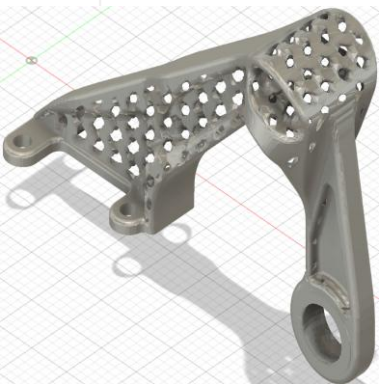
Topology optimization simulations



Von-Mises Stress distribution for the Topology Optimized Model

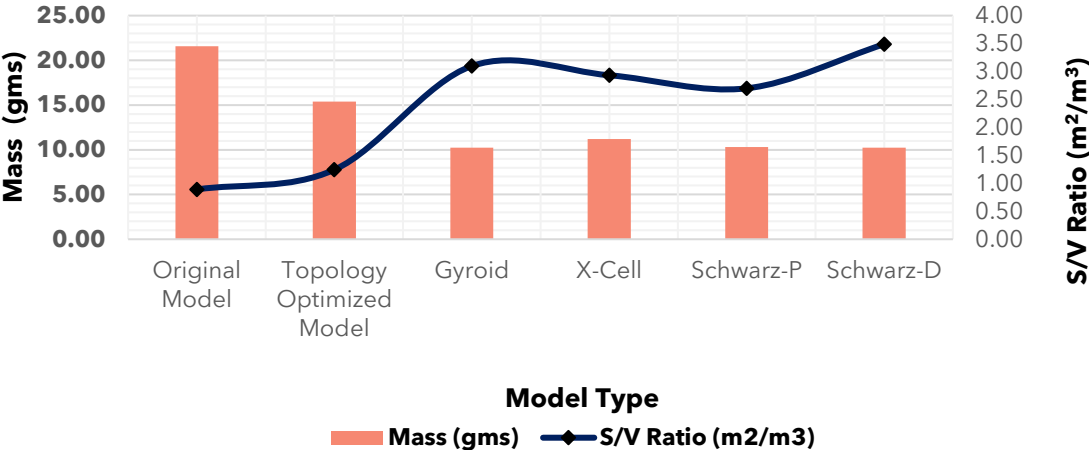


Max. VM Stress~**1072 MPa**
~20% reduction in Max. VM Stress

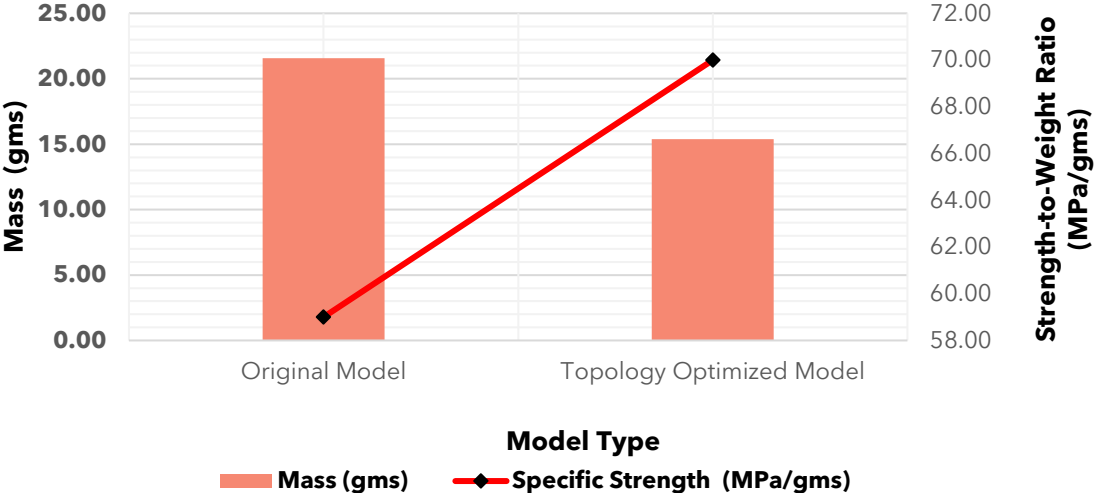


Comparative evaluation of bracket design approaches

Surface-to-Volume ratio of all Bracket design models



Specific Strength of all Bracket models



Property	Original Model	Topology Optimized Model
Mass (gms)	21.58	15.39
S/V Ratio (m²/m³)	0.89	1.24
% Mass Reduction	0	29
Specific Strength (MPa/gms)	59	70 (19% Increased)

Lattice structures			
Gyroid	X-Cell	Schwarz-P	Schwarz-D
10.24	11.22	10.31	10.23
3.10	2.93	2.70	3.49
53	48	52	53

Conclusion

- ❑ Designed and analyzed an aircraft bracket using shape, topology, and lattice optimization methods.
- ❑ Achieved up to **53% mass reduction** with lattice structures (Gyroid, Schwarz-D) and **29% with Topology optimization**.
- ❑ Topology optimization achieved the **highest specific strength of 70 MPa/g**, representing a **19% improvement** over the original bracket model.
- ❑ **Lattice models enhanced surface-to-volume ratio (~74%)**, promoting lightweight and functional integration.
- ❑ Established a **simulation-to-manufacturing workflow** for bearing bracket applications and experimental validation.



Thank You

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